

INTRUDER ALERT SYSTEM IN SMART HOME BASED ON IOT TECHNIQUE

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Abstract— The aim of the proposed work is to identify the moving human beings in a smart home security system. In the conventional home monitoring arrangement, Closed-Circuit Television (CCTV) is used which exactly record the videos without giving any feedback to the house owner if any suspected person there. Therefore, there is a need of moving object detection system and warning system for the safety of senior citizen in the home. The proposed hardware architecture is carried out using Arduino and Raspberry Pi 3 processor. The Raspberry Pi 3 process the images received through the Passive Infra Red (PIR) sensor and webcam assembled on Raspberry Pi 3. The PIR sensor activates the webcam, if any movement is detected around the sensor. The captured image is compared with two frames, if any motion is detected immediate alert in the form of call and mail is shared to the house owner in addition to the detected the motion object image.

Keywords— PIR Sensor, Internet of Things, Raspbery Pi, Smart Home, sensor, Closed-Circuit Television

INTRODUCTION

House is a place where the people along with senior citizen live and store wealth and asset of their entire life. Hence, there is a need of monitoring the house without any intruders. In traditional pattern, the security system is based on CCTV device supervise the different surroundings namely house, office buildings, rooms, accident road etc. [1]. CCTV will continuously monitor the situation for 24 hours and store the details in the storage space whenever the owner is present or not in the home [2]. Even though there are many benefit in the CCTV monitoring system, there exit some drawback also. CCTV monitoring system just continuously monitor the surrounding environment and don't sent any alert and warning if any intruders trying to enter in the place used. This type of monitoring system use huge data storage space and bandwidth because of endless video streaming. The great advantage of Internet of Things (IoT) is share the monitored data continuously to the destination with least human interference.

Nowadays, IOT pays a vital role in the smart city application, smart lighting, telecommunication, health monitoring in the hospital. Many research are under process in the field of IOT in smart communication. A smart security

monitoring system was developed by Charadva et al., (2014) [3] based on IOT technology. These system has feedback mechanism to the house owner for intimating the intruders in house with the use of PIR sensor, glass break detector and door open sensor. An innovative IOT based smart alert system in home is introduced by Tanwar et al. (2017) [4]. The smart monitoring system was developed using Raspberry Pi and Infrared sensor. This benefit of this module is used to give alert to the owners with least delay through email.

Bangali et al. (2013) proposed smart home security system in accordance with GSM technology [2]. These modules uses web camera for continuously monitor the motion object and gives warning to the user though sound using bazzar. In addition, sms is sent to the user with the help of GSM-GPS Module. Smart Surveillance Security system connected to mobile phone was developed by Chuimurkar et al. (2016) using PIR Sensor and Raspbery PI processor [5]. This monitoring system has the capability to sense the moving object and smoke and intimate the preventive measures against the intruders and fire. The system is incorporated with sound alert system when there is smoke or intruders in the house.

The intruder detection system was adopted using machine learning technology also. The advanced real nature intruder system namely Hybrid Intrusion Detection (HID) system was proposed by Faisal Alghayadh et al. (2021) [6]. Here, machine learning technique is adapted for intruder detection using different algorithm namely K-nearest neighbors, decision tree, random forest and misuse detection technique. The hybrid model namely convolution neural network (CNN) and bidirectional long short-term memory (BiLSTM) is adopted in any smart home gateway to detect anomalies intruders [7]. BiLSTM is used to store the data's with respect to time and CNN extract the features from the captured images.

In case of ensemble mode intrusion detection model, decision tree, Bayes, logistic regression classifiers are used for analyzing the performance [8]. The performance of the proposed system was examined by CICIDS2017 dataset.

From the above literature survey, the following technology was followed namely, home security system, motion detection. However, there is a lack of alert system using gmail when the intruder entered into the smart home security system using IoT technology.

II Proposed System Design

In system design section, the architecture of the proposed system is explained in details as in Fig. 1.

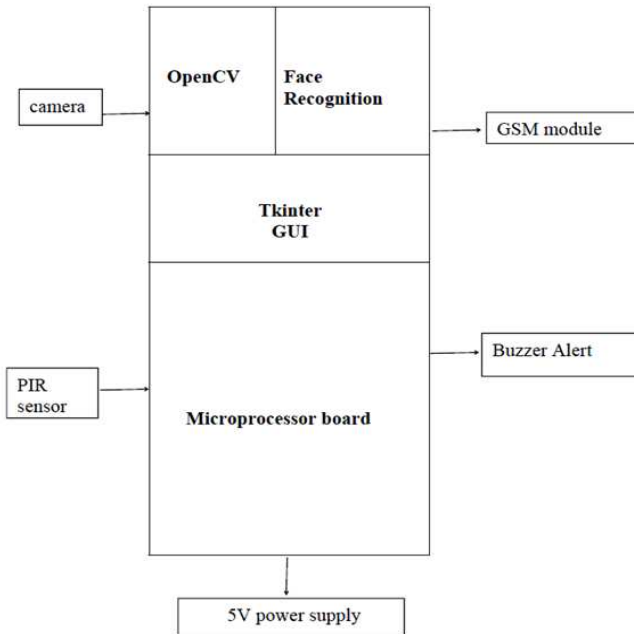


Fig. 1 System architecture

The PIR sensor is located in front of the door to detect any movement of human in the range about 5-7. When the PIR sensor continuously monitors the movement, webcam is activated. The webcam capture the image in the front door where the warning details are needed by the house owners. The captured image is stored in directory, analyzed and processed in microseconds. Histogram of Oriented Gradients (HOG) is used to extract the features in the captured image. The captured images from the webcam are compared with the images already stored in the database using K- nearest neighbor algorithm. In addition the data reduction for screening is done using map reduce algorithm. The Raspberry pi is act as a mini computer and act as vital element in the proposed surveillance system to manage all the function. The camera will capture the image when the person is moving in and around the camera. If any human presence is detected, an alert is given to the house owner through buzzer, call intimation and photo of the person is shared to the owner through gmail. The proposed system gives high security, since it is monitored by a single person. The furthermore advantage of the system is simple circuit to manage. The Operating System used to manage all the function is controlled using Raspbian OS. The flow diagram of the working system is shown in Fig. 2.

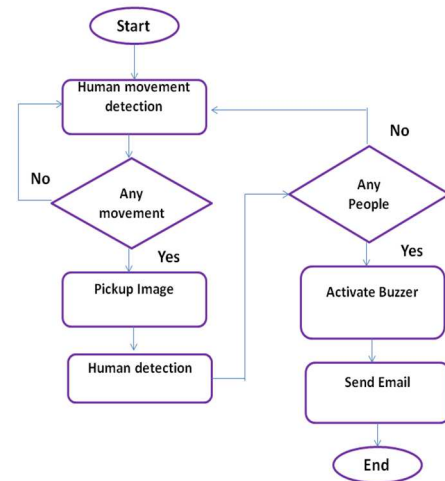


Fig. 2 Flow diagram of surveillance security system

III Hardware Design

The hardware design outline of the proposed system is explained in the following section as shown in Fig. 3. Over here, the different electronic components namely processor, sensors, interfaces and consolidation of all the components is discussed.

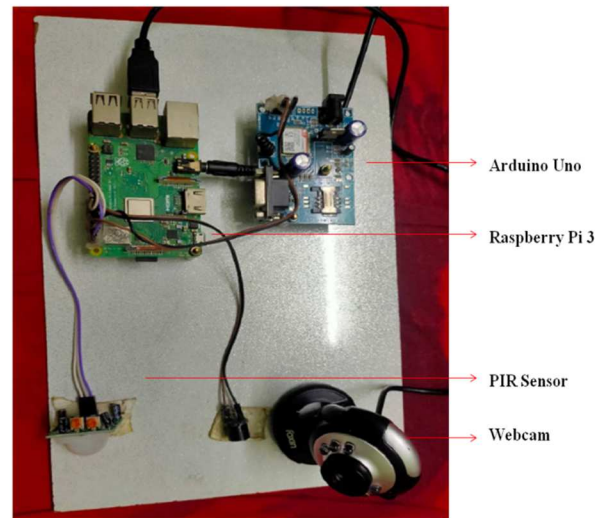


Fig. 3 Hardware architecture

The Raspberry Pi 3 module is used to coordinate the entire system. Raspberry Pi 3 is embedded with wireless LAN for wireless communication. The PIR sensor signals are gathered by Arduino uno using jumper cable. The Raspberry Pi 3 and Arduino uno are interfaced through USB cable. The Raspberry Pi 3 is assembled with webcam to capture the moving picture via USB cable. The warning is released to the house owner using buzzer module which is mounted to the Raspberry Pi 3 with the interfacing of GPIO port. The email notification is given to the user by connecting Raspberry Pi 3 to the internet.

IV System Setup

The function of entire electronic components present in the proposed system is explained in the following section.

(i) PIR Sensor on Arduino Uno

Arduino IDE app is used to program the PIR sensor to detect the motion of the object. The source code for this application is embedded in the Arduino uno by connecting it to the computer via USB cable.

(ii) Raspberry Pi 3

Raspbian Stretch operating system is used in Raspberry Pi 3. It is a freely available OS present in the website of Raspberry site . In Android smartphones, MicroSD is commonly used with the tenth version. Python program to used to run and process HOG and K- nearest neighbor algorithm.

(iii) Camera

Fswebcam library is used to run and process the webcam camera. Low resolution images are processed to improve the processing speed of the images. The processing of the images are within one microseconds.

(iv) Alarm

Alarm function can be made instantaneously when gpiozero pin interface is activated.

(v) E-mail

The emails are shared to the user, when some configurations made in the SMTP. The sender and receiver email address is configured and initialized for activating e-mail address. The configuration is done in the sender and receiver gmail account for allowing proper communication. Once the configuration is done, the email is send to the sender and recipient's without any issues.

VI Result and Discussion

In this section, monitoring system implementation and the result analyzed in the proposed system are discussed. This section discusses the implementation of IoT security monitoring system and the evaluation results. The camera capture and analyze the data even for a micro sec to achieve high security in home. In front of the door, camera and PIR sensors are placed to monitor the movement and to capture image. The monitoring system will be activated when the intruder approaches near the door. The security system activate the phone call if any intruder movement there. In addition the photo of the intruder has been sent to the mail.

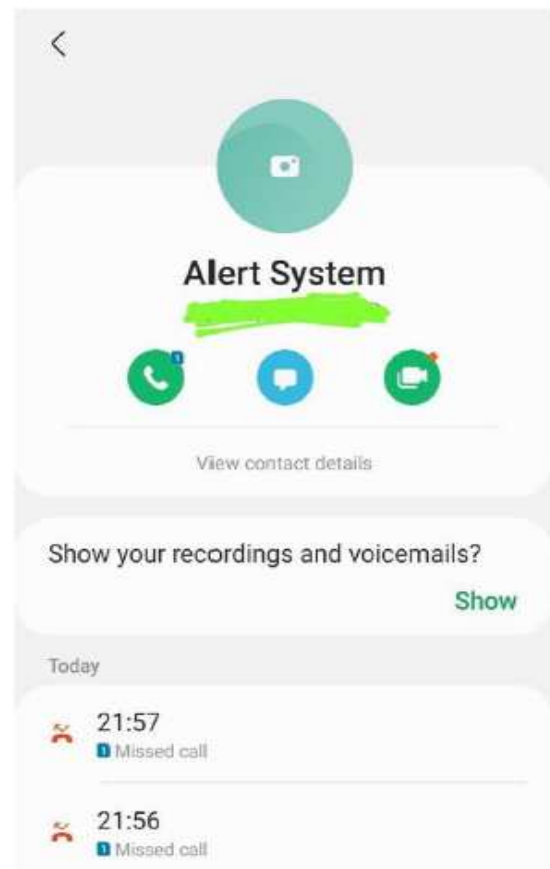


Fig. 4 Call alert System

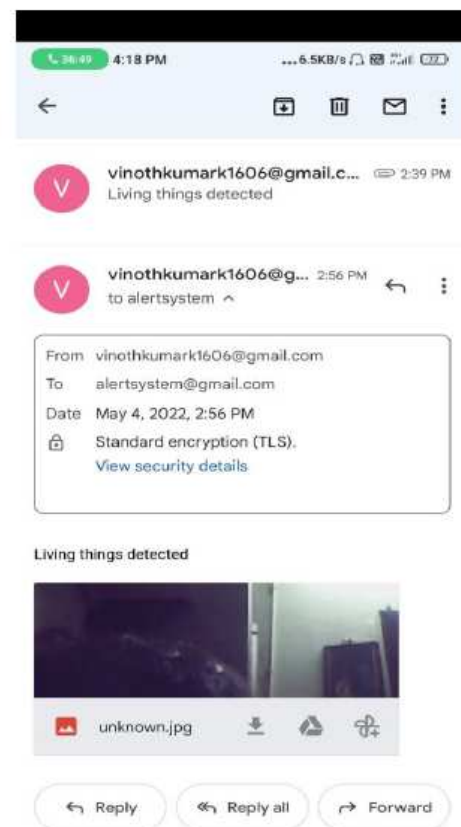


Fig. 5 Email alert System

IV CONCLUSION

The proposed work is a intruders monitoring system based on IoT technique. The system integrate many electronic components namely Arduino uno, buzzer, Raspberry Pi 3, webcam, interfaces and PIR sensor. The innovation of the entire intruder detection system is the ability to detect the human movement using HOG and K-nearest neighboring algorithm. The execution result shows that the system can detect an intruder within micro seconds. In future the accuracy of the entire system can be improved by adding additional feature extraction and classification methods.

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